04/06/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 05** **Time: 180 Mins.**

Answers

1. (4)	37. (3)	73. (2)	109. (2)	145. (4)
2. (1)	38. (2)	74. (3)	110. (3)	146. (2)
3. (1)	39. (4)	75. (3)	111. (4)	147. (1)
4. (4)	40. (4)	76. (1)	112. (4)	148. (4)
5. (4)	41. (3)	77. (3)	113. (1)	149. (2)
6. (2)	42. (4)	78. (1)	114. (4)	150. (4)
7. (2)	43. (4)	79. (2)	115. (3)	151. (2)
8. (1)	44. (3)	80. (2)	116. (3)	152. (1)
9. (3)	45. (1)	81. (3)	117. (2)	153. (2)
10. (4)	46. (2)	82. (3)	118. (2)	154. (4)
11. (4)	47. (1)	83. (4)	119. (3)	155. (4)
12. (2)	48. (3)	84. (1)	120. (2)	156. (4)
13. (3)	49. (3)	85. (2)	121. (2)	157. (2)
14. (2)	50. (4)	86. (2)	122. (2)	158. (4)
15. (1)	51. (2)	87. (3)	123. (4)	159. (4)
16. (1)	52. (2)	88. (3)	124. (2)	160. (3)
17. (4)	53. (1)	89. (1)	125. (1)	161. (3)
18. (4)	54. (3)	90. (2)	126. (3)	162. (2)
19. (2)	55. (2)	91. (2)	127. (2)	163. (1)
20. (2)	56. (1)	92. (3)	128. (3)	164. (1)
21. (4)	57. (4)	93. (3)	129. (3)	165. (1)
22. (1)	58. (4)	94. (2)	130. (4)	166. (3)
23. (3)	59. (4)	95. (3)	131. (4)	167. (3)
24. (2)	60. (3)	96. (4)	132. (2)	168. (3)
25. (2)	61. (4)	97. (2)	133. (3)	169. (2)
26. (1)	62. (2)	98. (2)	134. (1)	170. (4)
27. (2)	63. (3)	99. (2)	135. (4)	171. (4)
28. (1)	64. (2)	100. (3)	136. (1)	172. (4)
29. (4)	65. (4)	101. (3)	137. (1)	173. (2)
30. (1)	66. (2)	102. (2)	138. (2)	174. (4)
31. (4)	67. (3)	103. (3)	139. (4)	175. (1)
32. (3)	68. (3)	104. (2)	140. (4)	176. (1)
33. (1)	69. (2)	105. (3)	141. (4)	177. (4)
34. (3)	70. (1)	106. (4)	142. (2)	178. (2)
35. (4)	71. (4)	107. (4)	143. (2)	179. (1)
36. (4)	72. (4)	108. (3)	144. (4)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
Energy $= \frac{1}{2}LI^2$
$\left[\mathsf{ML}^2\mathsf{T}^{-2}\right] = L\left[A^2\right] \Rightarrow L = \left[\mathsf{ML}^2\mathsf{T}^{-2}A^{-2}\right]$

(2) **Answer :** (1)
Solution:
$a = 6v^3$
$v\frac{dv}{dx} = 6v^3$
$\int_2^v \frac{dv}{v^2} = \int_0^2 6dx$
$\left[\frac{v^{-2+1}}{-2+1}\right]_2^v = 6(x)_0^2$
$\Rightarrow \left[\frac{-1}{v}\right]_2^v = 6(2-0)$
$\Rightarrow -\left[\frac{1}{v} - \frac{1}{2}\right] = 12$
$\Rightarrow \frac{1}{v} - \frac{1}{2} = -12 \Rightarrow \frac{1}{v} = -12 + \frac{1}{2}$
$\Rightarrow v = -\frac{2}{23}$ m/s

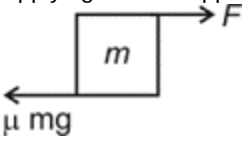
(3) **Answer :** (1)
Solution:
$x = y^2 + 2y + 2$
$\frac{dx}{dt} = 2y\frac{dy}{dt} + 2\frac{dy}{dt}$
$\frac{d^2x}{dt^2} = 2\left[y\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^2\right] + 2\frac{d^2y}{dt^2}$
Given, $a_y = 0$
$a_x = \frac{d^2x}{dt^2} = 2\left(\frac{dy}{dt}\right)^2$
$= 2(25) = 50$ m/s^2

(4) **Answer :** (4)
Solution:
By conservation of momentum
$m_1u_1 = (m_1 + m_2)v$
$\frac{m_1u_1}{(m_1+m_2)} = v$
$v = \frac{0.01\times400}{(0.01+1.2)} = \frac{4}{1.21}\,\mathrm{m/s}$
By work-energy theorem:
$\Delta K = W_f$ [$W_f \rightarrow$ work done by friction]
$0 - \frac{(m_1+m_2)V^2}{2} = -f_kd$
$\frac{(m_1+m_2)V^2}{2} = \mu_k\left(m_1 + m_2\right)gd$
$d = \frac{v^2}{2\mu_k g} = \frac{(4/1.21)^2}{2\times0.25\times10}$
$d \simeq 2.2\,\mathrm{m}.$

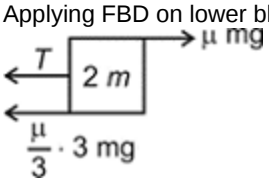
(5) **Answer :** (4)
Solution:
There is relative motion between both the blocks.

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Applying FBD on upper block



Applying FBD on lower block



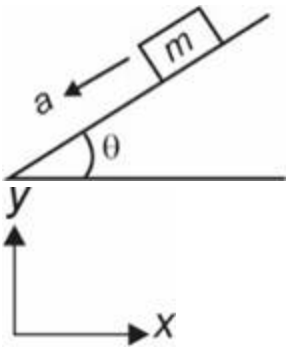
$T + \frac{\mu}{3} \times 3\text{mg} = \mu\text{mg}$

$\therefore T = 0$

In other cases also, Tension will be zero.

(6) **Answer :** (2)

Solution:



$a = g\sin\theta \;\; a_x = g\sin\theta\cos\theta \;\; a_x = \frac{g}{2}\sin 2\theta \;\; a_x$ will be maximum at $2\theta = 90^\circ$

i.e. $\theta = 45^\circ$

(7) **Answer :** (2)

(8) **Answer :** (1)

Solution:

At the highest point, speed

$v = 100\cos 60° = 50$ m/s horizontally

Conservation of linear momentum.

$2 \times 50\hat{i} = 1 \times 100\hat{j} + 1 \times \vec{v}_2$

$\Rightarrow \vec{v}_2 = 100\hat{i} - 100\hat{j}$

$\left|\vec{v}_2\right| = 100\sqrt{2}$ m/s

So kinetic energy of this part.

$= \frac{1}{2} \times 1 \times \left(100\sqrt{2}\right)^2 = 10^4 \text{J}$

(9) **Answer :** (3)

Solution:

For head on elastic collision of two identical bodies.

$m_1 = m_2 = m$ velocity gets exchanged

(10) **Answer :** (4)

Solution:

MOI about tangent to rim and lying in the plane of the disc = $\frac{5}{4}MR^2$

MOI about tangent to rim and perpendicular to plane of disc = $\frac{3}{2}MR^2$

(11) **Answer :** (4)

Solution:

A rigid body is said to be in mechanical equilibrium, if its linear momentum and angular momentum are not changing with time.

(12) **Answer :** (2)

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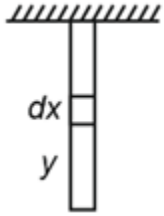
Hint:
If 'g' is acceleration due to gravity on the surface of earth, then

$\frac{4}{100}g = \frac{gR^2}{(R+h)^2}$ & $solve\ \ for$ 'h'

$\Rightarrow h = 4R$

(13) **Answer :** (3)
Solution:
NCERT Reference: Physics, Class-XI, Page No.-141

(14) **Answer :** (2)
Solution:



$dl = \frac{\left(\frac{M}{L}\right)xgdx}{AY}$

$l = \int dl = \frac{Mg}{ALY}\int_0^L xdx$

$= \frac{MgL}{2AY}$

$= \frac{WL}{2AY}$

(15) **Answer :** (1)

(16) **Answer :** (1)
Solution:

$x = vt = \sqrt{2g\overline{D}}\sqrt{\frac{2(H-D)}{g}}$

(17) **Answer :** (4)
Solution:

$\frac{A_2}{A_1} = \frac{T_2^4}{T_1^4} = \frac{(327+273)^4}{(273+27)^4} = \frac{6^4}{3^4} = \frac{16}{1}$

(18) **Answer :** (4)
Solution:

$P_0 = \rho\left(\frac{RT_0}{M}\right)$ and $4P_0 = \frac{\rho' \times R(3T_0)}{M}$

$4 = \frac{\rho'}{\rho} \times 3 \rightarrow \rho' = \frac{4}{3}\rho$

(19) **Answer :** (2)
Solution:

$\because \ \ \eta\left(Efficiency\right) = \frac{W}{Q_1}$

Q_1 ⇒ Supplied energy which should be positive.

$Q_{A\rightarrow B} = nC_V\,\Delta T + W$

$Q_{A\rightarrow B} = n\left(\frac{3R}{2}\right)\Delta T + 0$

$Q_{A\rightarrow B} = \frac{3}{2}nR\Delta T$

$= \frac{3}{2}[3P_0V_0 - P_0V_0]$

$Q\ A \rightarrow B = 3P_0V_0$

$Q\ B \rightarrow c = \frac{3}{2}[6P_0V_0 - 3P_0V_0] + 3P_0V_0$

$Q_{B\rightarrow C} = \frac{15}{2}P_0V_0$

$\because \ \ Q_1 = Q_{A\rightarrow B} + Q_{B\rightarrow C}$

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$Q_1 = 3P_0V_0 + \frac{15}{2}P_0V_0$

$Q_1 = \frac{21P_0V_0}{2}$

$\because W_{Cyclic}$ = [Area]$_{ABCDA}$

= $(3P_0 - P_0)(2V_0 - V_0)$

= $2P_0V_0$

$\therefore \ \eta = \frac{W}{Q_1}$

$\eta = \frac{4}{21}$.

(20) Answer : (2)

Solution:

$\gamma = 1 + \frac{2}{f}$

(21) Answer : (4)

Solution:

$y = kt^2 = \frac{1}{2}at^2$

$\frac{1}{2}at^2 = kt^2$

$a = 2k$,

$a = 4$ m/s^2

$\frac{T_1}{T_2} = \frac{2\pi\sqrt{\frac{l}{g}}}{2\pi\sqrt{\frac{l}{g+a}}} = \sqrt{\frac{g+a}{g}} = \sqrt{\frac{14}{10}}$

$\frac{T_1^2}{T_2^2} = \frac{14}{10} = \frac{7}{5}$

(22) Answer : (1)

Solution:

$l_2 - l_1 = \frac{\lambda}{2}$

$48 - 15 = \frac{\lambda}{2}$

$\lambda = 0.33 \times 2$ cm

$\frac{v}{f} = 0.66$ m

$v = 500 \times 0.66 = 330$ m/s.

(23) Answer : (3)

Solution:

From the given equation , $K = 0.2\pi$;

$\frac{2\pi}{\lambda} = 0.2\pi \Rightarrow \lambda = 10$ cm

$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = \frac{2\pi}{10} \times 2 = \frac{2\pi}{5} = 72^\circ$

(24) Answer : (2)

Hint:

Electric field due to uniformly charged sphere at its surface, $E = \frac{kq}{r^2}$

Solution:

$E = \frac{4kq}{d^2}$

$6 \times 10^6 = \frac{4 \times 9 \times 10^9 q}{16}$

$\frac{24}{9} \times 10^{-3} = q$

$q = \frac{8}{3} \times 10^{-3} C$

(25) Answer : (2)

Solution:

By theory of conductors

(26) Answer : (1)

Solution:

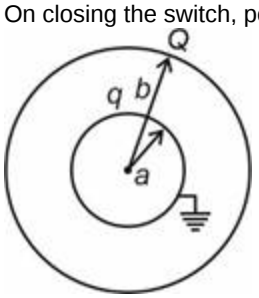
By energy conserve

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$\frac{1}{2}mv^2 = \frac{KZe^2}{r}$

$\Rightarrow \quad r = \frac{Ze^2}{2\pi\varepsilon_0 mv^2}$

(27) Answer : (2)
Solution:
On closing the switch, potential of inner sphere becomes zero.



Thus,
$V_{in} = 0$

$\frac{kq}{a} + \frac{kQ}{b} = 0$

$a = \frac{-Qa}{b}$

Change in charge when switch is closed = $q_f - q_i$

$\Delta Q = -Q\left(\frac{a+b}{b}\right)$

(28) Answer : (1)
Solution:
The given circuit is a balanced Wheatstone bridge.

$\Rightarrow \ R_{eq} = 10 \ \Omega \Rightarrow \ i = \frac{5}{10} = 0.5$ A

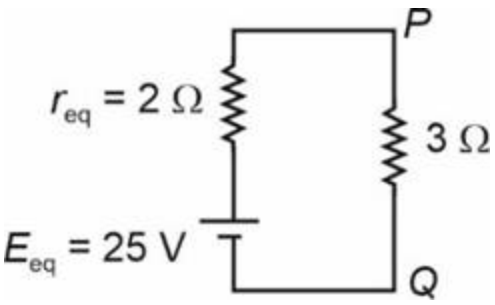
(29) Answer : (4)
Solution:
Treating 6 Ω as internal resistance of 15 V cell and 3 Ω as internal resistance of 30 V cell, then both cells are in parallel connection across *PQ*

$\therefore \ E_{eq} = \frac{15\times3+30\times6}{6+3}$

$= \frac{45+180}{9} = \frac{225}{9} = 25$ V

And $r_{eq} = \frac{6\times3}{6+3} = 2\ \Omega$

$\Rightarrow$



$\Rightarrow \ I_P = \frac{25}{3+2} = 5$ A

(30) Answer : (1)
Solution:
NCERT Reference: XI, Page No. 165

(31) Answer : (4)
Solution:
Effective Length = $\frac{\lambda}{2}$

Force = $i\, l_{eff}\, B = i \times \frac{\lambda}{2} B$

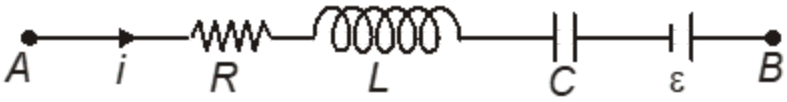
(32) Answer : (3)

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Solution:
For diamagnetic materials, $\mu_d = 0$

(33) Answer : (1)
Solution:
$\phi = \vec{B} \cdot \vec{A} = B_0 \left(2\hat{i} + 3\hat{j} + 4\hat{k}\right) \cdot \left(Lb\hat{k}\right)$
$\phi = 4B_0 Lb$ Wb

(34) Answer : (3)
Solution:



$V_A - IR - L\frac{di}{dt} - \frac{q}{C} + \varepsilon - V_B = 0$
$10 - 1 - 1 \times 10^{-3} \times 10^3 - \frac{q}{1 \times 10^{-6}} + 6 = 0$
q = 14 μC

(35) Answer : (4)
Solution:
$E = -M\frac{dI}{dt}$
$E = M\ I_o\ w \sin wt$
$E_{\max} = M\ I_o w = \ 0.001 \times 10 \times 100\pi$
$E_{\max} = \pi\ volts$

(36) Answer : (4)
Solution:
$R = 10\ \Omega$
$X_L = 0.2 \times 200 = 40\ \Omega$
$\cos\left(\theta\right) = \frac{1}{\sqrt{2}} = \frac{R}{Z}$
$\Rightarrow \quad |(X_L - X_C)| = R$
$\Rightarrow \quad (40 - X_C) = 10 \ \Rightarrow \ X_C \ = 30\ \Omega$
or $\quad X_C - 40 = 10 \ \Rightarrow \ X_C = 50\ \Omega$

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(37) Answer : (3)
Solution:
$B_0 = \frac{E_0}{c} = \frac{6}{3 \times 10^8} = 2 \times 10^{-8}\mathrm{T}$
Propagation direction = $\widehat{E} \times \widehat{B}$
$\hat{i} = \hat{j} \times \widehat{B}$
$\Rightarrow \ \widehat{B} = \hat{k}$

(38) Answer : (2)
Hint:
$v = \frac{c}{\mu}$
Solution:
$n_1 \sin\theta_1 = n_2 \sin\theta_2$
$n_1 \sin 30^\circ = 1 \ \ \sin 90^\circ$
n_1 = 2
Now, $\ v = \frac{c}{n_1} = \frac{3 \times 10^8}{2}$
v = 1.5 × 10^8 m/s

(39) Answer : (4)
Solution:
$E = -13.6 \times \frac{Z^2}{n^2}$ eV
For hydrogen atom $\ \Rightarrow \quad E = \frac{-13.6 \times 1}{n^2}$ eV ...(i)
For He^+ ion $\Rightarrow \quad E' = \frac{-13.6 \times 2^2}{n^2}$ eV ...(ii)

$\frac{E}{E'} = \frac{1}{4} \Rightarrow E' = 4E$

(40) Answer : (4)
Solution:



$\frac{1}{f} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{R} + \frac{1}{R}\right) = \frac{1}{R}$
Focal length of water lens.
$\frac{1}{f_w} = \left(\frac{4}{3} - 1\right)\left[\frac{1}{-R} - \frac{1}{R}\right] = -\frac{2}{3R}$
Focal length of combination
$\frac{1}{F} = \frac{1}{R} - \frac{2}{3R} + \frac{1}{R} = \frac{3-2+3}{3R}$
$F = \frac{3R}{4} = \frac{3f}{4}$

(41) Answer : (3)
Solution:
By theory.

(42) Answer : (4)
Hint:
Observation of photoelectric effect experiment.

Energy of incident photon	=	Maximum kinetic energy acquired by electron	+	Work function

Solution:
Due to energy loss inside the metal, photoelectrons emitted from a surface have different speeds.

(43) Answer : (4)
Solution:
Both mass number and proton number are conserved.

(44) Answer : (3)
Solution:
When junction is forward biased its potential barrier and depletion width, both decreases.

(45) Answer : (1)
Solution:
$\overline{A+B} = \overline{A} \cdot \overline{B}$
$\overline{A \cdot B} = \overline{A} + \overline{B}$
$\overline{\overline{A} + \overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}}$
= $A \cdot B$
$\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}}$
= $A + B$

CHEMISTRY

(46) Answer : (2)
Solution:
$n_{urea} = \frac{15}{60} = 0.25$
$n_{water} = \frac{90}{18} = 5$

$x_{urea} = \frac{0.25}{5.25} = 0.0476 \simeq 0.05$

(47) Answer : (1)
Solution:
The number of degenerate orbitals in 3rd shell of H-atom is 9.

(48) Answer : (3)
Solution:
For isoelectronic species, ionic radii decreases as the atomic number increases.

(49) Answer : (3)
Solution:

	IUPAC name	IUPAC official name
a.	Unnilunium	Mendelevium
b.	Unnilquadium	Rutherfordium
c.	Unnilseptium	Bohrium
d.	Unniltrium	Lawrencium

(50) Answer : (4)
Solution:
90° bond angles in SF_6 are 12.

(51) Answer : (2)
Solution:
Percentage ionic character = $\frac{\mu_{observed} \times 100}{\mu_{calculated}}$
$\mu_{observed} = 2.5 \times 3.33 \times 10^{-30}$ C m
$\mu_{calculated} = 1.6 \times 10^{-19} \times 125 \times 10^{-12}$ C m
% ionic character
$= \frac{2.5 \times 3.33 \times 10^{-30} \times 100}{1.6 \times 125 \times 10^{-31}} = 41.6\%$

(52) Answer : (2)
Solution:
$\Delta S = 2.303 \ \ nR \ \log \frac{V_2}{V_1}$
$= 2.303 \times 1 \times 2 \log \frac{20}{2}$
$= 2.303 \times 2 \times \log 10 = 4.606 \ calK^{-1}mol^{-1}$

(53) Answer : (1)
Solution:
At equilibrium $\Delta G = 0$
For both reversible and irreversible processes for an ideal gas $\Delta U = 0$

(54) Answer : (3)
Solution:
$pH = 7 + \frac{1}{2}(pK_a + logc)$
$= 7 + \frac{1}{2}(4.7 + \log 0.1) = 7 + \frac{1}{2}(4.7 - 1) = 8.85$

(55) Answer : (2)
Hint:
For any gaseous equilibrium
$K_P = K_C(RT)^{\Delta ng}$
Solution:
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $K_P = K_C$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $K_P = K_C(RT)^{-2}$
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_P = K_C(RT)$
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $K_P = K_C(RT)^{-1}$

(56) Answer : (1)

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Solution:
The oxidation number of Cl in ClO_2^- is +3 which is the intermediate oxidation state for Cl and it can be oxidised and reduced simultaneously.

(57) Answer : (4)
Solution:
$n_{factor} = \frac{2\times1\times5}{5+1} = \frac{10}{6} = \frac{5}{3}$

$E = \frac{71\times3}{5} = \frac{213}{5} = 42.6\ g$

(58) Answer : (4)
Hint:
CO forms a stable complex with haemoglobin, this found to be poisonous.
Solution:
Zeolites are widely used as a catalyst in petrochemical industries for cracking of hydrocarbons and isomerisation.
The highly poisonous nature of CO arises due to its ability to form a complex with haemoglobin which is about 300 times more stable than the oxygen-haemoglobin complex.
Being heavy and non-supporter of combustion CO_2 is used as fire extinguisher.

(59) Answer : (4)
Hint:
Percentage of nitrogen = $\frac{28\times \text{volume of } N_2 \text{ gas at STP}\times100}{22400\times \text{mass of organic compound}}$
Solution:
V_1 = 30 mL
P_1 = 715 – 15 = 700 mm Hg
T_1 = 300 K

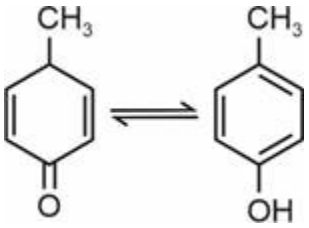
So, $V_2 = \frac{P_1\times V_1\times273}{T_1\times760} = \frac{700\times30\times273}{300\times760}$

= 25.1 mL

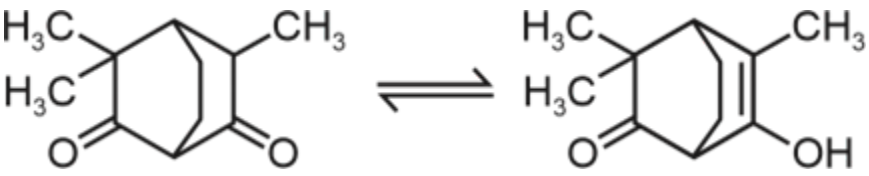
Percentage of nitrogen = $\frac{28\times25.1\times100}{22400\times0.1}$

= 31.4%

(60) Answer : (3)
Hint:
Bridgehead carbon cannot be sp^2 hybridized
Solution:
•



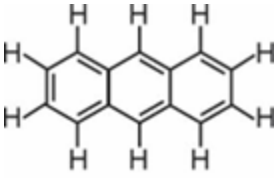
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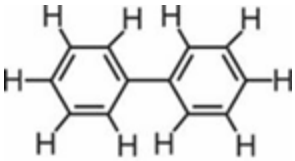
(61) Answer : (4)
Solution:

Compounds	Number of σ-bonds
(a)	26

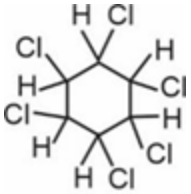
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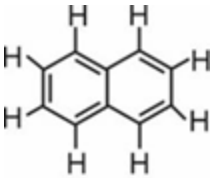
(Anthracene)

(b)  23

(Biphenyl)

(c)  18

(Gammaxane)

(d)  19

(Naphthalene)

(62) **Answer :** (2)
Solution:
Cyclic, planar species having (4n + 2)π electrons in conjugation are aromatic in nature.

(63) **Answer :** (3)
Solution:
According to Le Chatelier's principle solubility of gases in liquids decreases with rise in temperature.

(64) **Answer :** (2)
Solution:
$X_A = \frac{2}{3},\ X_B = \frac{1}{3}$
$P_T = X_A P_A^o + X_B P_B^o$
$= \frac{2}{3} \times 300 + \frac{1}{3} \times 200 = \frac{800}{3}\ mm\ Hg$
$Y_B = \frac{P_B}{P_{\text{Total}}} = \frac{\frac{1}{3} \times 200}{\frac{800}{3}} = 0.25$

(65) **Answer :** (4)
Solution:
Cathode: $\mathrm{Cu}^{2+}(\mathrm{aq}) + 2\mathrm{e}^- \rightarrow \mathrm{Cu}(\mathrm{s})$
Cu^{2+} is a stronger oxidising agent than H^+, hence will get reduced and form Cu at cathode.

(66) **Answer :** (2)
Solution:

Cell	**Reaction at Anode**
Mercury cell	– $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$
Dry cell	– $Zn(s) \rightarrow Zn^{2+} + 2e^-$
H_2–O_2 fuel cell	– $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$
Lead storage battery (on discharging)	–

$$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$$

(67) Answer : (3)
Hint:

Rate of reaction: $-\frac{1}{5}\frac{d[Br^-]}{dt} = \frac{1}{3}\frac{d[Br_2]}{dt}$

Solution:

$\frac{d[Br_2]}{dt} = \frac{3}{5}\left(-\frac{d[Br^-]}{dt}\right)$

= $\frac{3}{5} \times 4.5 \times 10^{-3}$

= 2.7×10^{-3} mol L^{-1} sec^{-1}

(68) Answer : (3)
Solution:

$k = \frac{2.303}{70}\log_{10}\left(\frac{100}{20}\right) = \frac{2.303}{t}\log_{10}\left(\frac{100}{10}\right)$

⇒ t = 100 minutes

(69) Answer : (2)
Solution:

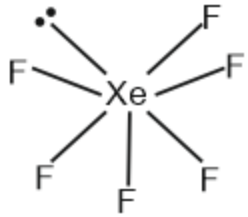
Metaphosphoric acid $(HPO_3)_n$ – $H_3PO_3 + Br_2$, heated in a sealed tube.

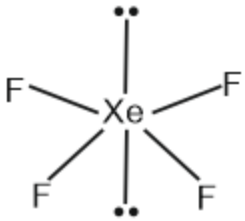
Phosphonic acid – $P_2O_3 + H_2O$

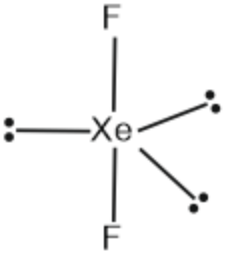
Hypophosphoric acid – Red P_4 + alkali

Pyrophosphorous acid – $PCl_3 + H_3PO_3$

(70) Answer : (1)
Solution:

XeF_6  1 L.P

XeF_4  2 L.P

XeF_2  3 L.P

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(71) Answer : (4)

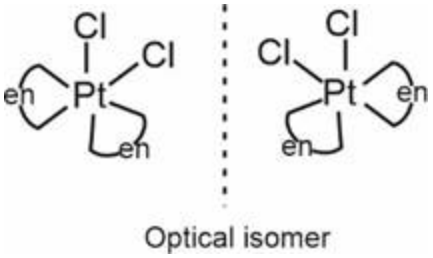
Solution:

Ion	Number of unpaired electron(s)	Spin only magnetic moment value in BM
Ni^{2+}	2	2.83
Cr^{2+}	4	4.90
Ti^{3+}	1	1.73
Co^{2+}	3	3.87

(72) Answer : (4)
Solution:

Ion	Ionic radii (pm)
Nd^{3+}	99
Eu^{3+}	95
Dy^{3+}	91
Yb^{3+}	86

(73) Answer : (2)
Solution:



Optical isomer

cis-form of the complex exist as d, l pair.

(74) Answer : (3)
Solution:

Co^{3+} (high spin)

d^6 configuration = $t_{2g}^4 e_g^2$ (unsymmetrical)

Fe^{3+} (low spin)

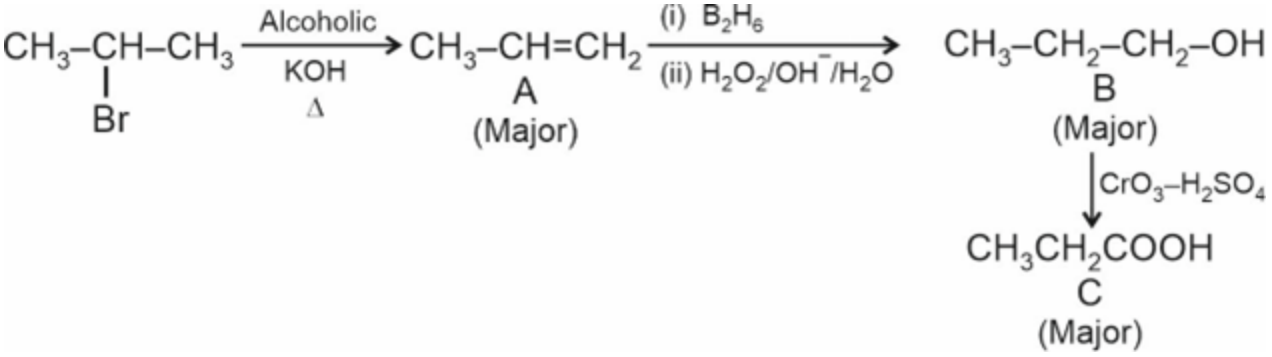
d^5 configuration = $t_{2g}^5 e_g^0$ (unsymmetrical)

so these complexes show Jahn Teller distortion.

(75) Answer : (3)
Solution:

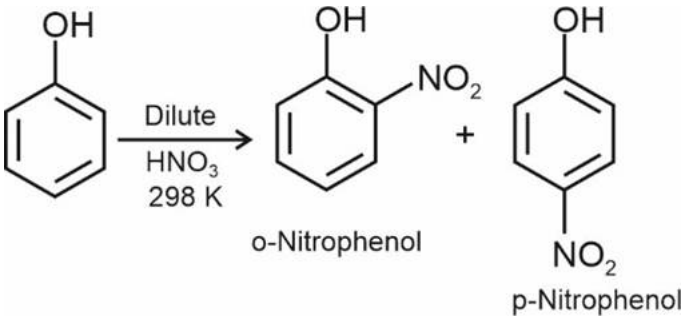
Stability of carbocations decides the increasing order of reactivity towards S_N1 reaction.

(76) Answer : (1)
Solution:

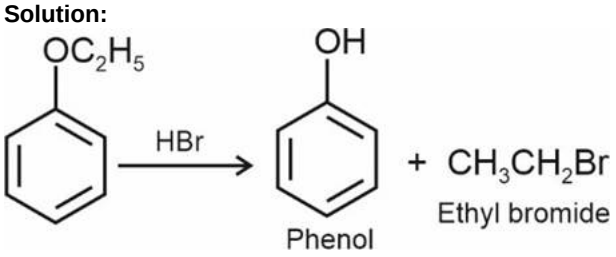


(77) Answer : (3)
Solution:

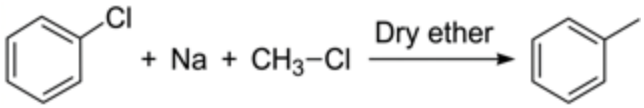
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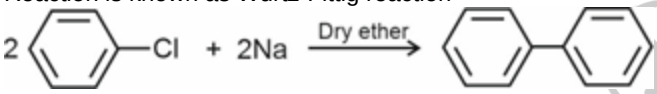
(78) Answer : (1)
Solution:



(79) Answer : (2)
Solution:



Reaction is known as Wurtz-Fittig reaction



Reaction is known as Fittig reaction

$CH_3 - Br + AgF \rightarrow CH_3 - F$

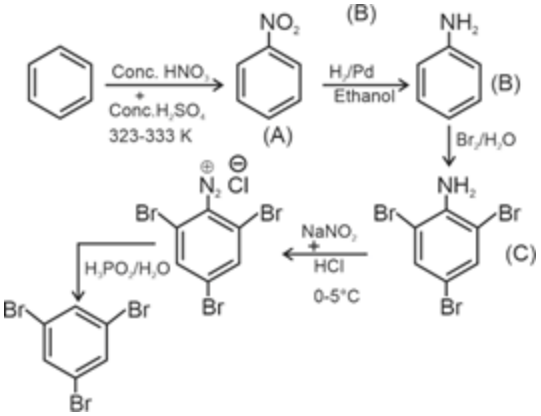
Reaction is known as Swarts reaction

$CH_3\,Cl + NaI \xrightarrow[\text{acetone}]{\text{dry}} CH_3\,I + NaCl$

Reaction is known as Finkelstein reaction

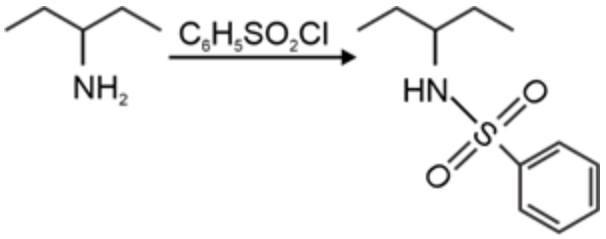
(80) Answer : (2)
Solution:
Aldehydes which do not contain an α-hydrogen atom, undergo self-oxidation and reduction (disproportionation) reaction on treatment with concentrated alkali.

(81) Answer : (3)
Solution:



(82) Answer : (3)
Solution:

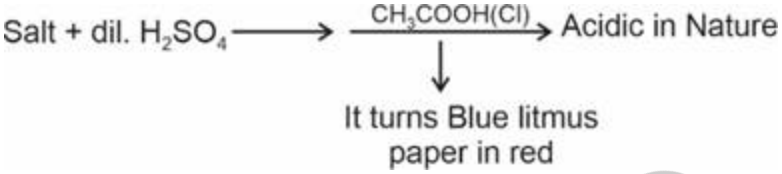
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(83) Answer : (4)
Solution:

(Vitamins)	(Deficiency Diseases)
Thiamine	– Beri-beri
Riboflavin	– Cheilosis
Pyridoxine	– Convulsion
Ascorbic acid	– Scurvy

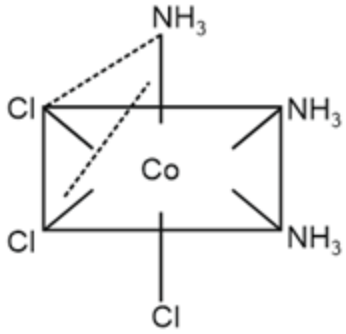
(84) Answer : (1)
Solution:



So, correct Answer is : Acetate, CH_3COO^-

(85) Answer : (2)
Solution:
Ba^{2+} is group V cation while Ni^{2+} is group IV cation.
Group reagent for group V – $(NH_4)_2CO_3$ in presence of NH_4OH.
Group reagent for group IV – H_2S in presence of NH_4OH.
Brown ring test is the confirmatory test of NO_3^-ion.

(86) Answer : (2)
Solution:
(a) $Ni(PPh_3)_4$ is tetrahedral complex and does not show steroisomerism.
(b) Number of triangular faces (containing one N and two Cl atoms) is fac form of $[Co(NH_3)_3Cl_3]$ are 3



(c) $[Co(en)_2(ox)]Cl$ have total 2 steroisomers while $[Co(en)_2Cl_2]Br$ has total 6 possible isomers.

(87) Answer : (3)
Solution:
$C_xH_y(g) + \left(x + \frac{y}{4}\right)O_2(g) \rightarrow xCO_2(g) + \frac{y}{2}H_2O(l)$
5 mL $\quad 5\left(x + \frac{y}{4}\right) \quad\quad 5x$
Volume of CO_2 produced = 25 – 15 = 10 = 5x
x = 2
Volume of O_2 consumed $= 15 = 5\left(x + \frac{y}{4}\right)$

$3 = 2 + \frac{y}{4}$

y = 4

So, the molecular formula of hydrocarbon is C_2H_4.

(88) Answer : (3)

Solution:

Energy absorbed by electron 13.6 × 2 = 27.2 eV

Energy required for ionisation = 13.6 eV

$\therefore$ KE = 27.2 – 13.6 = 13.6 eV = 13.6 × 1.6 × 10^{-19} = 2.17 × 10^{-18} J

$KE = \frac{1}{2}\ mv^2$

$V = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2\times 2.17\times 10^{-18}}{9.1\times 10^{-31}}}$ = 2.19 ×10^6

$\lambda = \frac{h}{mv}$

$\therefore \quad \lambda = \frac{6\times 10^{-34}}{9\times 10^{-31}\times 2.19\times 10^{6}} = 3.04\times 10^{-10}$m

(89) Answer : (1)

Solution:

$H_2S \rightleftharpoons H^+ + HS^-$, $\quad K_1 \frac{[H^+][HS]}{[H_2S]}$

$HS^- \rightleftharpoons H^+ + S^{2-}$, $\quad K_2 \frac{[H^+][S^{2-}]}{[HS]}$

K = $K_1 \times K_2$ = 1.3 × 10^{-20}

Ksp = $[M^{2+}]\ [S^{2-}]$ = 6 × 10^{-21}

$S^{2-} = \frac{6\times 10^{-21}}{0.046} = 1.3\times 10^{-19}$

$K = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$

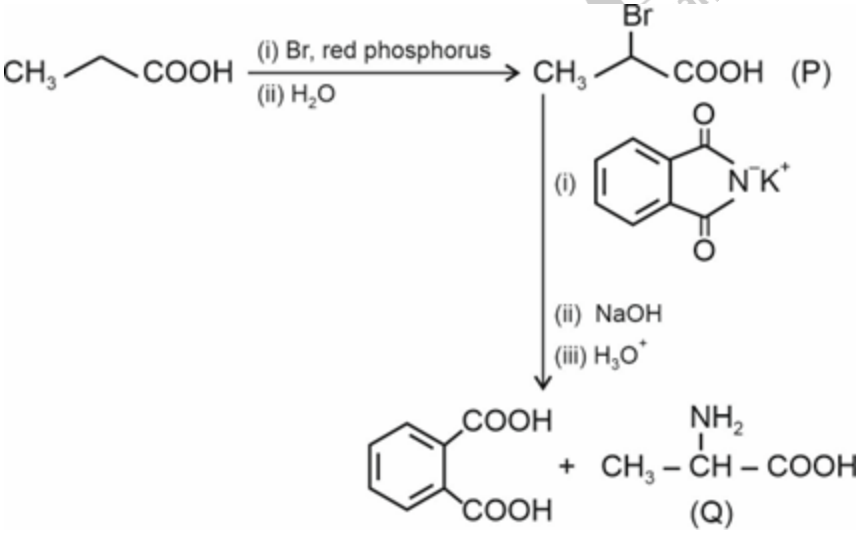
$1.3\times 10^{-20} = \frac{[H^+]^2\times 1.3\times 10^{-19}}{0.1}$

$\Rightarrow \quad [H^+] = \sqrt{\frac{1.3\times 10^{-10}\times 0.1}{1.3\times 10^{-19}}} = 10^{-1}$

pH = –log$[H^+]$ =1

(90) Answer : (2)

Solution:



BIOLOGY

(91) Answer : (2)

Solution:

Name of genus and species are printed in italics..Name of the author appears after the specific epithet, i.e., at the end of the biological name and is written in an abbreviated form, e.g., *Mangifera indica* Linn.

(92) Answer : (3)
Hint:
Trypanosoma causes sleeping sickness.
Solution:
• The parasitic form of flagellated protozoan causes disease like sleeping sickness.
• Amoeboid protozoans produce false feet to capture prey.
• Ciliated protozoans have cavity that opens to the outside of cell surface.
• Sporozoans include diverse organisms that have infectious spore-like stage in their life cycle.

(93) Answer : (3)
Solution:
Cyanobacteria lack flagella throughout life cycle.

(94) Answer : (2)
Solution:
Both the member of group ascomycetes and deuteromycetes reproduce asexually through conidia.

(95) Answer : (3)
Solution:
In pteridophyte , the gametophyte is found in cool, damp, shady places and sex organs antheridia and archegonia are produced on the gametophyte.

(96) Answer : (4)
Solution:
Fucus is diploid, shows oogamous reproduction.
Polytrichum is a moss and *Riccia* is a liverwort. In angiosperms, ovules are covered by ovary.

(97) Answer : (2)
Solution:
In *Laminaria*, plant body is differentiated in frond, stipe and hold fast.

(98) Answer : (2)
Solution:
Petiole allows the leaf blade to flutter in wind, thereby cooling the leaf and bringing fresh air to leaf surface.

(99) Answer : (2)
Hint:
Calyx is the outermost whorl of a flower.
Solution:
Sepals are green, leaf-like and protect the flower in the bud stage. In vexillary aestivation, the largest petal overlaps the two lateral petals which in turn overlap the two smallest anterior petals(keel). Solanaceae is widely distributed in tropics, sub tropics and even temperate zones.

(100) Answer : (3)
Hint:
'A' is the hypodermis of monocot stem.
Solution:
'C' in the given figure is phloem of monocot stem and it lacks phloem parenchyma.

(101) Answer : (3)
Solution:
Guard cells are dumb-bell shaped in monocots and have chloroplast.

(102) Answer : (2)
Solution:
Secondary growth does not occur in leaves.
Vascular bundles in leaves are conjoint and closed.

(103) Answer : (3)
Solution:
Cilia is a surface structure in which 9 + 2 arrangement of microtubules are found. Cilia is involved in cell motility.

(104) Answer : (2)
Solution:
The three layers of the bacterial envelopes are glycocalyx, cell wall and plasma membrane.
Plasma membrane is structurally similar in prokaryotes and eukaryotes.

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(105) Answer : (3)

Solution:

DNA content in a meiocyte at G_2 phase is 40 pg. This means the DNA content in diploid cell is 20 pg. The haploid cells formed after meiosis II will have DNA content of 10 pg.

(106) Answer : (4)

Solution:

Neuroglial cells support and protect neurons.

(107) Answer : (4)

Solution:

The birth canal is formed by the cervical canal (the passage through the cervix) and the vagina.

(108) Answer : (3)

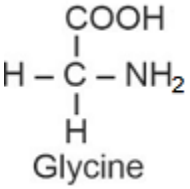
Solution:

If the product is at the high energy level than the substrate, then the reaction is endothermic.

(109) Answer : (2)

Solution:

Neutral amino acids have equal number of amino group(s) and carboxylic group(s).



Glycine

Tyrosine is an aromatic amino acid.

(110) Answer : (3)

Hint:

This sugar is abundant in corn and grapes.

Solution:

Fructose is six carbon sugar with keto group. Ribose and deoxyribose are five carbon sugars with aldehyde group.

(111) Answer : (4)

Solution:

When dried tissue is fully burnt, the carbon compounds are oxidised to gaseous forms like CO_2 and water vapour and the remnant is called ‘ash’. This ash contains many inorganic elements like calcium, magnesium *etc*.

(112) Answer : (4)

Hint:

Part of fatty acids

Solution:

Glycerol is simple lipid which is trihydroxy propane. Many lipids have both glycerol and fatty acids. Here the fatty acids are found esterified with glycerol. They can be then monoglycerides, diglycerides and triglycerides.

(113) Answer : (1)

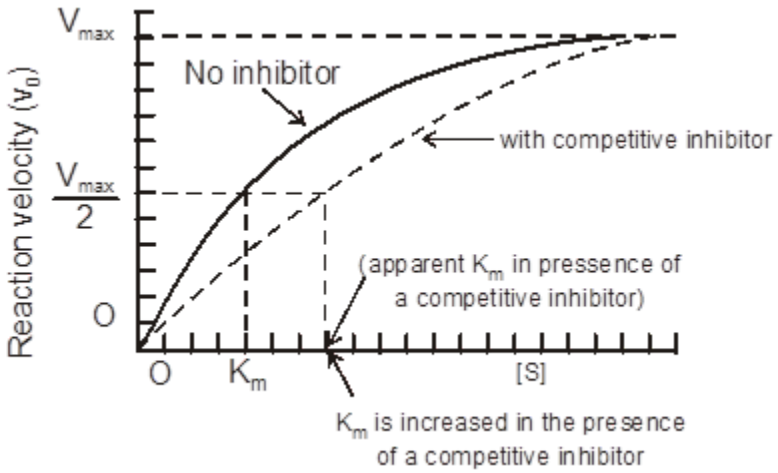
Hint:

Malonate is a competitive inhibitor.

Solution:

In competitive inhibition, by increasing the concentration of substrate, inhibition can be reversed.

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$\therefore$ V_{max} is constant and K_m increases.

(114) Answer : (4)

Solution:

Every 100 mL of oxygenated blood delivers about 5 mL O_2 to tissues under normal physiological conditions but can deliver up to 15 mL O_2 to tissues during strenuous exercise.

(115) Answer : (3)

Solution:

By counting the number of QRS-complexes that occur in a given time period, one can determine the heart rate of an individual.

(116) Answer : (3)

Solution:

The beginning of diplotene is recognised by the dissolution of synaptonemal complex and the recombined homologous chromosome separate except at the site of crossovers. These X shaped structures are called chiasmata.

(117) Answer : (2)

Solution:

Prophase is marked by the initiation of condensation of chromosomal material.

(118) Answer : (2)

Hint:

In human beings, the membrane of erythrocyte has 52% proteins and 40% lipids.

Solution:

The integral proteins run throughout the lipid bilayer. while peripheral proteins are found on the surface of membrane only.

(119) Answer : (3)

Solution:

Since the splitting of the water molecule takes place on the inner side of the thylakoid membrane, hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids.

(120) Answer : (2)

Hint:

Grana are absent in bundle sheath cells.

Solution:

Both bundle sheath cells and mesophyll cells are involved in CO_2 fixation.

(121) Answer : (2)

Solution:

Apical dominance is inhibited by cytokinin.

(122) Answer : (2)

Solution:

When fats are used in respiration, the RQ is less than 1. Calculations for a fatty acid, tripalmitin, if used as a substrate is shown below:

$$\underset{\text{Tripalmitin}}{2(C_{51}H_{98}O_6)} + 145\,O_2 \rightarrow 102\,CO_2 + 98H_2O + \text{energy}$$

$$RQ = \frac{102\,CO_2}{145O_2} = 0.7$$

(123) Answer : (4)

Solution:

CoA is used during the conversion of α-ketoglutaric acid into succinyl coenzyme A in citric acid cycle.

(124) Answer : (2)

Solution:

In pineapple, flowering and fruit set up is synchronized by ethylene and it increases female flowers in cucumber.

(125) Answer : (1)

Solution:

In *Citrus*, embryo formed by nucellus and integuments are genetically similar. Therefore seeds are apomictic in nature. Being a dicot, *Citrus* seeds lack coleorhiza. Embryo formed by zygote, integument and nucellus are diploid.

(126) Answer : (3)

Hint:

Remains of nucellus are found in perispermic seeds.

Solution:

Beet and black pepper are perispermic seeds having remains of nucellus.

(127) Answer : (2)

Hint:

In angiosperms seeds have better adaptive strategies for dispersal to new habitat.

Solution:

Advantages of seed to angiosperms are

- To have food reserve for nourishment of young seedlings.
- Presence of hard seed coat provide protection to young embryo.
- Help the species to colonise in other areas.

(128) Answer : (3)

Solution:

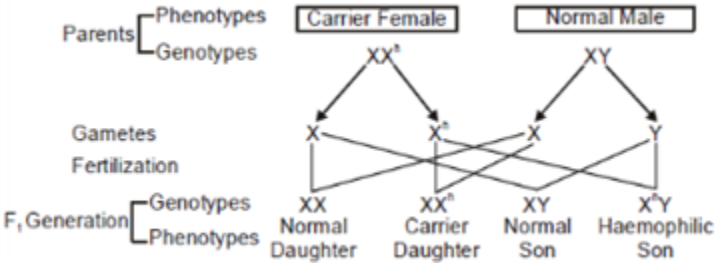
A × AB

$I^A I^O \times I^A I^B$

Cannot produce 'B' blood group with genotype $I^B I^B$.

(129) Answer : (3)

Solution:



(130) Answer : (4)

Solution:

Types of gamete = 2^n

$\because$ Genotype of organisms is AaBb

$\therefore$ n = 2 $\Rightarrow 2^2$ = 4 types of gametes.

(131) Answer : (4)

Hint:

Tidal volume is the air breathed in or out in a normal breath.

Solution:

Respiratory Volumes	Values
Tidal Volume (TV)	500 mL
Inspiratory Reserve Volume (IRV)	2500–3000 mL
Expiratory Reserve Volume (ERV)	1000–1100 mL
Residual Volume (RV)	1100–1200 mL

(132) Answer : (2)

Hint:
Work oppositely
Solution:
As urine gets filled in the bladder, the stretch receptors present on the walls of the bladder send signals to the CNS.
The CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine.

(133) Answer : (3)
Hint:
ADH prevents diuresis
Solution:
ADH/vasopressin acts at kidney and stimulates reabsorption of water and Na^+ from latter parts of the distal tubules. It reduces loss of water through urine.
An excessive loss of fluid from the body can activate osmoreceptors which stimulate the hypothalamus to release vasopressin from the neurohypophysis.

(134) Answer : (1)
Solution:
A specialised centre present in the medulla region of the brain called respiratory rhythm centre is primarily responsible for this regulation. Another centre present in the pons region of the brain called pneumotaxic centre can moderate the functions of the respiratory rhythm centre. Neural signal from this centre can reduce the duration of inspiration and thereby alter the respiratory rate.

(135) Answer : (4)
Solution:
Pivot joint is present between radius and ulna just below the elbow.

(136) Answer : (1)
Solution:



Increase in Ca^{+2} level leads to the binding of calcium with a subunit of troponin on actin filaments which thereby removes the masking of active sites for myosin.

(137) Answer : (1)
Solution:
Hypothalamus releases gonadotrophin releasing hormone/GnRH (X) which stimulates the pituitary to synthesise LH and FSH (Y).
• LH induces ovulation and also maintains the corpus luteum.
• Androgens are responsible for the development of secondary sexual characters in males.
• Progesterone acts on the mammary glands and stimulates the formation of alveoli.

(138) Answer : (2)
Hint:
Bulging of eyes is a characteristic of these patients.
Solution:
Symptoms of cretinism are given in the question. Hypersecretion of thyroxine causes exopthalmic goitre. Simple goitre and myxedema are also caused by deficiency of thyroxine.

(139) Answer : (4)
Solution:
Ctenoplana belongs to the phylum Ctenophora, in which fertilisation is external with indirect development.

(140) Answer : (4)
Solution:
– Cyclostomes have 6-15 pairs of gill slits, show anadromous migration and circular mouth without jaws is present.
– Cranium and vertebral column are cartilaginous.

(141) Answer : (4)
Solution:
Epididymis is male accessory duct.

(142) Answer : (2)
Solution:

Structure having corpora quadrigemina

Cerebral aqueduct is a canal which passes through midbrain and connects 3rd ventricle of diencephalon with 4th ventricle of medulla. Cavity of cerebrum is called paracoel.

(143) Answer : (2)
Solution:

Spermatheca are present in 6th abdominal segment which opens into the genital chamber. In a fertile female, spermatheca is found filled with spermatozoa, received during copulation.

(144) Answer : (4)
Hint:
True for testes.
Solution:
In female frogs, the ovaries are situated near kidneys and there is no functional connection with kidneys. Males have urinogenital ducts. Larva of frog is called tadpole.
In male frogs, a pair of yellowish ovoid testes are found adhered to the upper part of kidneys by a double fold of peritoneum called mesorchium.

(145) Answer : (4)
Hint:
Free living worm.
Solution:
Planaria is not an endoparasite but is found in streams and rivers.

(146) Answer : (2)
Hint:
Spermiation occurs after spermiogenesis.
Solution:
The spermatids are transformed into spermatozoa (sperms) by the process called spermiogenesis.
After spermiogenesis, sperm heads become embedded in the Sertoli cells and are finally released from the seminiferous tubules by the process called spermiation.

(147) Answer : (1)
Solution:
Bioreactors are large vessels for biomass culture.

(148) Answer : (4)
Solution:
A plasmid is autonomously replicating, circular and extra-chromosomal DNA. Some plasmids carry antibiotic-resistance genes.

(149) Answer : (2)
Solution:
Theca layer is formed around secondary follicle but it is organised into an outer theca externa layer and inner theca interna layer within tertiary follicle.

(150) Answer : (4)
Solution:
Administration of progesterone will maintain the endometrium and prevent it from shedding.

(151) Answer : (2)
Solution:
Foreign gene inserted in ampicillin resistant gene will cause insertional inactivation so all transformants will grow in tetracycline containing medium.

(152) Answer : (1)
Solution:
The DNA fragments get separated (resolve) according to their size through sieving effect provided by the agarose gel. Since DNA fragments are negatively charged molecules, they can be separated by forcing them to move towards the positive electrode/anode under an electric field through the gel matrix.

(153) Answer : (2)
Hint:
Vectorless gene transfer
Solution:
Electroporation, biolistics and chemical mediated gene transfer are included in direct methods of gene transfer. Disarmed retrovirus can be used as a vector to transfer functional ADA cDNA gene into the patient's lymphocytes for the treatment of SCID.

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(154) Answer : (4)
Solution:
The nematode parasite that infects the roots of tobacco plants is *Meloidogyne incognita.*

(155) Answer : (4)
Hint:
Identify a vitamin which is essential for proper vision.
Solution:
Golden rice is a transgenic variety of rice which contains good quantities of β-carotene, a principal source of vitamin A.
Vitamin K helps in blood clotting and vitamin D helps in calcium deposition in bone.

(156) Answer : (4)
Solution:
Distance between the two gene is inversely proportional to the frequency of recombination. The genes for white eye color and miniature wings were loosely linked and showed 37.2% recombinant types.

(157) Answer : (2)
Solution:
In humans, male is sex determiner. Male produces 50% sperms with X-chromosome and 50% sperms with Y-chromosome.
Hence, in each pregnancy there would be always 50% probability of a female child.

(158) Answer : (4)
Hint:
3' end of tRNA attaches with an amino acid.
Solution:
This tRNA has UAC in anticodon loop, which means it will bind to the codon AUG of mRNA.

(159) Answer : (4)
Solution:
Transformation occurs when one bacterium picks up free-floating DNA and incorporates it into its own genome.

(160) Answer : (3)
Solution:
Repressor protein of *lac* operon is synthesised by *lac i* gene and referred to as negative regulation.

(161) Answer : (3)
Solution:
Severo Ochoa enzyme is polynucleotide phosphorylase.

(162) Answer : (2)
Solution:
Deoxyadenosine is a nucleoside, which lacks phosphate group.

(163) Answer : (1)
Solution:
Alexander Fleming while working on *Staphylococci* bacteria, discovered the first antibiotic penicillin.

(164) Answer : (1)
Solution:
Monascus purpureus is used for the commercial production of statins.

(165) Answer : (1)
Hint:
Logistic growth shows sigmoid curve.
Solution:
Exponential growth results in a J-shaped curve. Exponential growth occurs when resources (food and space) are unlimited.
According to Darwin, when resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number.

(166) Answer : (3)
Solution:
Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

(167) Answer : (3)
Solution:
Net primary productivity is the available biomass for the consumption to heterotrophs. Primary productivity depends on the plant species inhabiting a particular area.

(168) Answer : (3)

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Solution:
Law of 10% is followed here

$$\underset{500\ \text{kJ}}{\text{Plant}} \rightarrow \underset{50\ \text{kJ}}{\text{Grasshopper}} \rightarrow \underset{5\ \text{kJ}}{\text{Frog}} \rightarrow \underset{0.5\ \text{kJ}}{\text{Snake}}$$

(169) Answer : (2)
Solution:
Steller's sea cow got extinct from the Russia.

(170) Answer : (4)
Solution:
Cichlid fish is an ecologically unique assemblage in lake Victoria in east Africa. They became prone to extinction due to alien species invasion.

(171) Answer : (4)
Solution:
Bacterium *Bacillus anthracis* is included in bio weapon.

(172) Answer : (4)
Solution:
The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid *Salmonella typhimurium.*
Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance.

(173) Answer : (2)
Solution:
This process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography is called adaptive radiation.

(174) Answer : (4)
Hint:
Cu^{2+} are released
Solution:
IUDs increase phagocytosis of sperms within the uterus and the Cu^{2+} released suppress sperm motility and the fertilising capacity of sperms. The hormone releasing IUDs, in addition, make the uterus unsuitable for implantation and the cervix hostile to the sperms.

(175) Answer : (1)
Solution:
They walked upright
Their fossils were first obtained from South Africa as indicated by word "*Australo*". Savannah Grasslands are found in Ethiopia and Tanzania.

(176) Answer : (1)
Solution:
Though options are many, all these ARTs require extremely high precision handling by specialised professionals and expensive instrumentation. Therefore, these facilities are presently available only in very few centres in the country. Obviously, their benefits is affordable to only a limited number of people. Emotional, religious and social factors are also deterrents in the adoption of these methods.

(177) Answer : (4)
Solution:
About 65 mya, the dinasaurs suddenly disappeared from the earth.

(178) Answer : (2)
Solution:
Morphine is extracted from the latex of poppy plant, *Papaver somniferum*.
It is a very effective sedative and painkiller and is very useful in patients who have undergone surgery.

(179) Answer : (1)
Solution:
Sporozoite is infective stage for human and gametocyte for mosquito.

(180) Answer : (1)
Solution:
Bone marrow and thymus are responsible for differentiation of immature lymphocytes into antigen-sensitive lymphocytes.

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